Literature Review

Flight Delay Prediction

Gradient Boosting, Temporal Evaluation, and Leakage-Resistant Features

Team T029

Section E • Machine Learning Project

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Abstract

This review examines machine-learning methods for predicting whether a flight arrives at least 15 minutes late, using the 2013 New York flights dataset covering JFK, LaGuardia (LGA), and Newark Liberty (EWR). It connects prior approaches to the project's implemented workflow: label construction, missing-target handling, pre-departure feature selection, chronological model evaluation, and leakage checks. The project notebook reports 336,776 original records; excluding 9,430 records without an observed arrival-delay label leaves 327,346 labelled flights, with a delayed-flight prevalence of 24.47% in the EDA output. Logistic Regression and gradient-boosted trees are considered within a time-ordered validation design. The review emphasizes that evaluation scores are meaningful only when feature availability, class prevalence, and the prediction cutoff are made explicit. No model-performance result is asserted here unless it is computed and reported from the project's executed evaluation.

Keywords: flight delay prediction; binary classification; Gradient Boosting; temporal validation; data leakage; class imbalance; aviation analytics.

Project in brief. This project uses historical flight records associated with New York's JFK, LaGuardia (LGA), and Newark Liberty (EWR) airports to predict whether a flight's arrival delay reaches 15 minutes. This review focuses on machine-learning approaches for structured flight data, predictor availability at prediction time, and evaluation on later observations rather than randomly mixed records.

1. Introduction

Flight delays affect passenger itineraries, airline operations, and airport scheduling. Predicting delay risk before a flight operates could support planning and decision-making. The task is challenging because delay outcomes may reflect interactions among airline schedules, airport conditions, routes, and time-dependent operational patterns.

The project uses the 2013 New York flights dataset, with 336,776 original records across 21 columns as described in the project materials. The EDA notebook reports 9,430 records with missing `arr_delay`; these records are excluded because the binary arrival-delay target is unknown, leaving 327,346 labelled flights (97.2% of the original records). The notebook reports that 24.47% of labelled flights meet the delay threshold. These counts and prevalence should be reproduced from the final modelling data and reported for each chronological split, because the target rate can change over time.

This is a binary-classification problem. The project investigates gradient boosting as its main Phase 1 model and compares it with simpler baselines. Predictions must rely on information that would be available at the declared prediction time; leakage control and chronological evaluation are therefore as important as algorithm choice.

2. Machine Learning Approaches

2.1 Interpretable Baselines

Logistic Regression is a useful baseline for structured classification. It estimates the probability of delay from a weighted combination of input features and provides a relatively straightforward reference for evaluating more complex models. Its limitation is that nonlinear relationships and interactions may require explicit feature construction.

Decision Trees can represent nonlinear splits, while Random Forests combine multiple trees to reduce dependence on any single tree. These are relevant comparisons for tabular flight data because airline, airport, schedule, and route characteristics may relate to delay risk in different ways. Reviews and benchmarking studies of flight-delay prediction provide broader context for comparing model families [2, 3]. Baseline comparison helps establish whether additional complexity produces a meaningful improvement.

2.2 Gradient Boosting

Gradient Boosting builds an ensemble of weak learners sequentially, with each new learner improving predictions according to a loss function. Tree-based boosting can represent nonlinear relationships and feature interactions without requiring every relationship to be specified manually. This is relevant to exploratory analysis of departure time, airline, route, and airline–hour interactions.

For example, the association between scheduled departure hour and delay risk may differ by airline. A tree-based model can learn such conditional patterns when supported by the data. Gradient Boosting is not guaranteed to outperform every alternative: results depend on feature quality, hyperparameters, and evaluation design. Learning rate, tree depth, and number of trees should be selected using validation data, not the final test period.

2.3 Temporal Deep Learning

Temporal models, including Long Short-Term Memory (LSTM) networks, can represent sequential dependencies when observations are organized into meaningful sequences. They may be useful

for modelling delay propagation, successive flights, or changing airport conditions. However, timestamps alone do not make a dataset a suitable sequence dataset. Sequence structure, prediction horizon, and information availability must be defined first.

Gradient boosting is therefore the primary focus of the current Phase 1 work. Temporal deep learning is a future extension unless valid sequential inputs can be constructed and evaluated under the same time-based protocol.

3. Exploratory Analysis and Feature Engineering

Exploratory analysis connects model design to the structure of the dataset. Relevant areas include missingness, time-of-day patterns, airline and route variation, and possible interactions between airline and scheduled hour. Observed associations should not be interpreted as proof of causation.

3.1 Scheduled and Categorical Features

Potential predictors include airline, origin, destination, scheduled departure time, calendar variables, and route distance, provided these fields exist and are available at prediction time. Categorical variables may require one-hot encoding or another suitable representation. Hour and calendar features can capture repeating patterns, while route-related variables can distinguish flights with different operating characteristics.

Cyclical encoding can represent repeating quantities such as hour of day or day of week using sine and cosine transformations. For a periodic variable x with period P:

$$x_{\sin} = \sin\left(\frac{2\pi x}{P}\right), \qquad x_{\cos} = \cos\left(\frac{2\pi x}{P}\right).$$

This representation places values near the end and beginning of a cycle close to one another.

3.2 Missing Data

Missing values require investigation before modelling. Their frequency and distribution should be checked by column and, where appropriate, by target class or time period. Removing every row with any missing value can unnecessarily discard useful observations. Imputation, exclusion, or missingness indicators should instead be chosen according to the meaning of each field and fitted without using validation or test information.

Flights without an observed arrival delay are excluded from the binary arrival-delay task because their target is unknown. The final report should state how missing values were handled and how many records remained after cleaning. The original dataset size should not be presented as the final modelling sample size unless the two are identical.

3.3 Data Leakage and Feature Availability

A predictor is valid only if it would be known when the prediction is issued. Actual arrival time, actual arrival delay, actual departure time, and other post-operation outcomes must not be used

as pre-departure predictors.

Leakage can also occur in historical aggregates. An airline's delay rate, for example, must be computed from outcomes observed before the flight being predicted, not from the complete dataset. Previous-flight information can be used only if it would already be available at the chosen prediction cutoff.

Congestion-related features require particular care. A count of scheduled flights from the same airport in the same hour may be defensible if it is calculated from schedule information assumed to be available at prediction time. The report should state this assumption explicitly. If a feature uses information unavailable at the cutoff, remove it or reconstruct it using an appropriate time cutoff. Future information can produce overly optimistic offline scores.

4. Project-Specific Data and Prediction Contract

The accompanying preprocessing notebook makes the prediction scenario explicit: estimate delay risk at 6 PM on the day before the scheduled flight. Under this contract, scheduled and calendar fields are candidate predictors, while realized operating outcomes are unavailable. The feature allow-list in the notebook includes carrier, origin, grouped destination, flight identifier, distance, scheduled departure and arrival minute representations, cyclical hour and calendar encodings, and `scheduled_origin_hour_count`.

The notebook explicitly excludes `dep_time`, `dep_delay`, `arr_time`, `air_time`, `arr_delay`, `id`, `name`, `tailnum`, the target label, and split metadata from the model inputs. This is an important safeguard: a field can be present in a historical dataset but still be invalid for a prediction made before departure. The flight identifier is used as a categorical field in the current allow-list, so its value should be justified and compared against a model without it; if it functions mainly as a near-unique identifier rather than a stable operational category, it may add noise or encourage memorization.

The scheduled origin-hour count also deserves a documented availability check. A count calculated from the complete schedule may be valid if the schedule is known by the stated cutoff and the feature uses no realized outcomes. It should not be described as a real-time congestion measurement: it counts scheduled flights, not actual aircraft or airport conditions. This distinction matters when interpreting the model's intended use.

5. Evaluation Design

5.1 Target and Class Prevalence

Let D_i denote the arrival delay in minutes for flight i. The binary target is

$$y_i = \begin{cases} 1, & D_i \geq 15, \\ 0, & D_i < 15. \end{cases}$$

The reported positive-class prevalence of approximately 24.47% means delayed flights form the smaller class in the reviewed dataset. Accuracy alone may hide weak detection of delayed flights,

so prevalence should be reported separately for training, validation, and test periods. These rates may change over time.

5.2 Chronological Validation

The project plan uses the following chronological partition:

Subset	Planned period	Role
Training	January–August	Fit preprocessing and model parameters.
Validation	September–October	Select model settings and the decision threshold.
Testing	November–December	Estimate performance on later observations.

These periods must match the actual dates and coverage of the dataset. Chronological splitting is appropriate for prediction on future flights because it reduces the risk of evaluating on records temporally mixed with training data. Rolling-origin validation can provide an additional robustness check across multiple chronological cutoffs.

Preprocessing steps such as imputers, scalers, and encoders should be fitted on training data only. Validation data should support model selection and threshold selection; the test period should be reserved for final evaluation.

5.3 Evaluation Metrics

The project considers PR-AUC, ROC-AUC, lift, and recall at a specified precision:

- **PR-AUC / Average Precision:** summarizes the precision–recall trade-off and is informative when the positive class is less prevalent.
- **ROC-AUC:** measures how well the model ranks delayed flights above non-delayed flights.
- **Recall at a specified precision:** measures the proportion of delayed flights detected while meeting a chosen precision requirement.
- **PR-AUC lift:** compares average precision with the positive-class prevalence baseline:

$$\text{Lift} = \frac{\text{Average Precision}}{\text{Positive-class prevalence}}.$$

When reporting lift, average precision and prevalence must come from the same evaluation subset. Results should include the baseline model, selected model, test-period prevalence, and threshold used for threshold-dependent metrics. No performance score should be claimed until measured on the held-out test period.

5.4 Reproducibility and Model Comparison

The implementation uses a fixed random seed and pipeline-based preprocessing, with transformations fitted on training data. The workflow compares a simple baseline with a more complex

candidate and selects model settings and a decision threshold using validation data. The held-out test period is intended for the final estimate only. The final report should include the number of rows and positive-class prevalence in each split, the model selected on validation data, the threshold-selection rule, and test metrics. If a model is refitted on training plus validation data after reporting the test result, that test period must not then be reused for further model selection.

6. Research Gap and Project Contribution

The project is motivated by the gap between building a classifier on historical records and estimating how it will behave on future flights. A model can exploit patterns unavailable at prediction time, and random splitting can obscure changes across time. The project's contribution is therefore not simply the use of gradient boosting, but the combination of a structured baseline, feature checks, chronological evaluation, and transparent metrics.

Three issues receive particular attention:

- **Temporal realism:** evaluate on later observations rather than relying only on randomly mixed records.
- **Leakage-resistant features:** verify that historical, congestion-related, and other derived variables use only information available at prediction time.
- **Changing prevalence and missing context:** compare delay rates across periods and acknowledge that the dataset does not contain direct weather observations.

Without direct weather observations, the model cannot directly learn the effect of measured storms or other weather conditions. Other recorded variables may reflect related patterns, but they are not substitutes for explicit weather data. The current binary task also excludes records with missing arrival delay, which commonly include cancelled or diverted flights; it therefore does not predict cancellation risk and should not be interpreted as covering every scheduled flight. Findings should not automatically be generalized to all airports, years, or operating environments.

7. Proposed Workflow

The project can be implemented as the following reproducible sequence:

1. **Audit the data:** verify provenance, column meanings, date coverage, duplicates, missingness, and target prevalence.
2. **Explore patterns:** summarize delay rates by airline, airport, scheduled hour, route, and time period.
3. **Build valid features:** encode categorical and scheduled-time information; verify historical and congestion features for future leakage.
4. **Establish a baseline:** train a simple classifier using the training period.

5. **Train gradient boosting:** select hyperparameters using validation data or rolling-origin folds, never the final test period.
6. **Evaluate on the test set:** report PR-AUC, ROC-AUC, lift, recall at the chosen precision, class prevalence, and limitations.

This workflow connects exploratory findings to model design and keeps development decisions separate from final evaluation.

8. Conclusion

The reviewed project data provide a basis for studying flight-delay prediction, but reliable conclusions depend on data preparation and evaluation. Gradient boosting is a reasonable primary model for structured flight records because it can learn nonlinear patterns and feature interactions. A simple baseline remains necessary to determine whether added complexity improves performance.

The most important safeguards are to use valid pre-departure features, check derived congestion and historical statistics for future leakage, preserve chronological order, and report class prevalence alongside model metrics. The lack of weather observations limits the explanatory context available to the model. Future work may investigate rolling-origin validation, additional verified external data, and temporal deep learning if appropriate sequential features can be constructed.

Summary at a Glance

- **Dataset:** 336,776 records and 21 columns as stated in the supplied review; reconfirm after final cleaning.
- **Goal:** predict arrival delay of at least 15 minutes for flights associated with JFK, LGA, and EWR.
- **Main model:** gradient boosting, compared with a simpler baseline.
- **Safeguards:** pre-departure-only predictors, leakage checks, and chronological train/validation/test periods.
- **Metrics:** PR-AUC, ROC-AUC, lift, and recall at a specified precision, with prevalence reported for each evaluation period.
- **Limitation:** no direct weather observations; derived congestion features require explicit availability assumptions.

Resources and References

The references below are carried over from the supplied review. Their metadata and links should be checked against the publisher or institutional source before final submission; this review does not treat the bibliographic details as independently verified.

[1] U.S. Department of Transportation, Bureau of Transportation Statistics, "On-Time Reporting." Available: https://www.bts.gov/topics/airlines-and-airports/number-14-time-reporting.

[2] S. Gharibzadeh, M. M. Pedram, M. Khademi, and A. Salajegheh, "Trends and gaps in machine learning based flight delay prediction in spatiotemporal and data-driven approaches: A systematic review," *Applied Soft Computing*, vol. 202, article 115761, 2026, doi: 10.1016/j.asoc.2026.115761.

[3] "Machine Learning Methods Benchmarking for Predicting Flight Delays: An Efficiency Meta-Analysis," *Sustainability*, vol. 17, no. 21, article 9887, 2025. Available: https://www.mdpi.com/2071-1050/17/21/9887.